

Top-k

it limits the model to choosing a token randomly from only the top-k tokens with the highest probability. For example, if k is set to 5, you are restricting the model to choose from only the top five tokens using the weighted random-sampling strategy. It worth mentioning that setting top-k to a higher number can help reduce repetitiveness, while setting top-k to 1 basically gives you greedy decoding.

Top-p

It restricts the model to randomly sampling tokens from a subset where their cumulative probabilities, starting from the highest and moving to the lowest, do not exceed p. For instance, if $p=0.35$, the tokens “learns,” “from,” and “student” are included because their probabilities of 0.2, 0.1, and 0.05, respectively, sum to 0.35. The model then uses the weighted random-sampling strategy to choose the next token.

Temperature

This parameter helps control the randomness of the model's output by altering the shape of the next-token probability distribution. Generally, increasing the temperature results in higher randomness, while decreasing it leads to lower randomness.